PTF ALL

Smoke or Fumes In Cabin Source Unknown - Fault Isolation

CORRECTIVE ACTION SUMMARY

Do Action Specified by Observed Fault Task
Replace Hydraulic Reservoir Pressurization Module
Replace Air Charging Valve Manifold Assembly
Correct Related Maintenance Messages
Clean/Replace Cabin Equipment
Replace Air Recirculation Filters
Replace Air Recirculation Fan

A. Initial Evaluation

CAUTION DO NOT REPLACE PARTS UNTIL YOU IDENTIFY, AND REMOVE THE SOURCE OF THE FUMES. THE FUMES CAN CAUSE THE CONTAMINATION OF A NEW PART. THEN THE NEW PART WILL BE THE SOURCE OF THE FUMES.

(1) Smoke or fumes from an external source can come into the air conditioning inlets.
   (a) The exhaust from other airplanes or ground equipment can cause smoke or fumes in the airplane.
   (b) Contaminated air from a ground cart can cause smoke or fumes in the airplane.
   (c) A fuel leak near the air conditioning inlets can cause smoke or fumes in the airplane.
   (d) The air distribution system can move the smoke or fumes into different areas of the airplane.

(2) Smoke or fumes can come from internal APU or engine problems that release oil into the pneumatic and/or air conditioning systems.

(3) Smoke or fumes can also come from the recirculation fans, gasper fans, equipment cooling fans, cabin temperature sensor fan, compressor assembly, IFE Equipment, materials in the cargo compartments, and hydraulic fluid from the reservoir pressurization module tubing.

(4) Smells in the cabin can come after an engine overhaul or compressor wash. This condition can occur when there is not a failure in the system. Put the airplane on watch as necessary.

(5) Smells in the cabin can occur again after the correction of a problem with a bleed air source (APU, engines). Additional flights or the replacement of ozone converters, trim air mufflers, heat exchangers, or cooling pack components may be necessary to remove all the contamination.

(6) Different sources can produce fumes with different smells. The list that follows has typical descriptions of the fumes from different sources to help you isolate the fault:
   (a) Oil fumes can cause mild irritation to the eyes and are described as:
      1) Sweat, locker room, or dirty sock
      2) Rancid cheese
      3) Wet dog
      4) Burning rubber
      5) Sour milk
   (b) Hydraulic fluid fumes can cause strong irritation to the eyes and are described as:
      1) Pungent
      2) Acrid
   (c) Fuel fumes are described as:
      1) Kerosene (burned)
      2) Kerosene (unburned)
      3) Acrid

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4) Bitter

(d) Electrical fault fumes are described as:
1) Acrid
2) Burning rubber
3) Sulfur

(e) Spills and unwanted material fumes can irritate the nose and throat and are described as:
1) Burning resin
2) Hot musty smell
3) Cooked chicken (bird strike)
4) Visible haze without odor

(7) Check the hydraulic system for an over-servicing condition before you continue. This is the task:
• Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

(8) Do these steps to simulate the conditions for the observed fault.
(a) Start the APU. This is the task:
• APU Starting and Operation, AMM TASK 49-11-00-860-801.

(b) Start the left and right engines. This is the task:
• Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
• Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

(d) Set these switches on the forward overhead panel to the specified positions:
1) ISOLATION VALVE switch to the OPEN position.
2) BLEED 1 and BLEED 2 switches to the ON position.
3) BLEED APU switch to the ON position.

(e) Pressurize the hydraulic reservoirs. This is the task:
• Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(f) Supply conditioned air with the left and right cooling packs. This is the task:
• Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(g) Set these switches on the forward overhead panel to the specified positions:
1) L RECIRC FAN and R RECIRC FAN switches to the OFF position.
2) WING ANTI-ICE switch to the OFF position.
3) ENG ANTI-ICE 1 and ENG ANTI-ICE 2 switches to the OFF position.

(9) If the smoke or fumes increase or if they stay after five minutes, then do these steps to isolate the source of the smoke or fumes:
(a) Operate the left and right packs with pneumatic air supplied by the APU.
1) Set these switches on the forward overhead panel to the specified positions:
a) BLEED 1 and BLEED 2 switches to the OFF position.
b) L PACK and R PACK switches to the AUTO position.
2) If the smoke or fumes increase or if they stay after five minutes, then do the Fault Isolation Procedure (APU) below.
a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
• Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
b) Stop the left and right engines. This is the task:
• Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
c) Stop the APU. This is the task:
• APU Usual Shutdown, AMM TASK 49-11-00-860-802.
3) If the smoke or fumes go away, then continue.

(b) Operate the left pack with pneumatic air supplied by engine 1.

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1) Set these switches on the forward overhead panel to the specified positions:
   a) BLEED APU switch to the OFF position.
   b) BLEED 1 switch to the ON position.
   c) R PACK switch to the OFF position.
2) If the smoke or fumes increase or if they stay after five minutes, then do the Fault Isolation Procedure (Engine 1) below.
   a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
   b) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
   c) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.
3) If the smoke or fumes go away, then continue.

(c) Operate the right pack with pneumatic air supplied by engine 2.
1) Set these switches on the forward overhead panel to the specified positions:
   a) BLEED 1 switch to the OFF position.
   b) BLEED 2 switch to the ON position.
   c) R PACK switch to the AUTO position.
   d) L PACK switch to the OFF position.
2) If the smoke or fumes increase or if they stay after five minutes, then do the Fault Isolation Procedure (Engine 2) below.
   a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
   b) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
   c) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.
3) If the smoke or fumes go away, then continue.

(10) If there are no smoke or fumes, then do these steps:
(a) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
(b) If the smoke or fumes increase or if they stay after five minutes, then do the Fault Isolation Procedure (Electrical Faults) below.
1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
2) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
3) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.

(c) If there are no smoke or fumes, then do these steps:

NOTE: Smoke or fumes that occurred in flight are not always possible to simulate on the ground.

1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
2) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
3) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.
4) Determine the source of the smoke or fumes by the description of the smell.
5) If you smelled oil fumes, then do the Fault Isolation Procedure (Oil Fumes) below.

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6) If you smelled hydraulic fluid fumes, then do the Fault Isolation Procedure (Hydraulic Fluid) below.
7) If you smelled fuel fumes, then do the Fault Isolation Procedure (Fuel Fumes) below.
8) If you smelled electrical faults, then do the Fault Isolation Procedure (Electrical Faults) below.
9) If you cannot find the source of smoke or fumes from the description, then there was an intermittent fault.

B. Fault Isolation Procedure (APU)

(1) Go to the FIM task for the observed fault: "Smoke or fumes in cabin: air supplied by APU" and do the steps in the task.
(a) Start the APU. This is the task:
• APU Starting and Operation, AMM TASK 49-11-00-860-801.
(b) Start the left and right engines. This is the task:
• Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.
(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
• Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.
(d) Set these switches on the forward overhead panel to the specified positions:
1) ISOLATION VALVE switch to the OPEN position.
2) BLEED 1 and BLEED 2 switches to the ON position.
3) BLEED APU switch to the ON position.
(e) Pressurize the hydraulic reservoirs. This is the task:
• Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.
(f) Supply conditioned air with the left and right cooling packs. This is the task:
• Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
(g) If the smoke or fumes go away, then you corrected the fault.
1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
• Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
2) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
3) Stop the left and right engines. This is the task:
• Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
4) Stop the APU. This is the task:
• APU Usual Shutdown, AMM TASK 49-11-00-860-802.

C. Fault Isolation Procedure (Engine 1)

(1) Go to the FIM task for the observed fault: "Smoke or fumes in cabin: air supplied by engine 1" and do the steps in the task.
(a) Start the APU. This is the task:
• APU Starting and Operation, AMM TASK 49-11-00-860-801.
(b) Start the left and right engines. This is the task:
• Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.
(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
• Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.
(d) Set these switches on the forward overhead panel to the specified positions:
1) ISOLATION VALVE switch to the OPEN position.
2) BLEED 1 and BLEED 2 switches to the ON position.
3) BLEED APU switch to the ON position.
(e) Pressurize the hydraulic reservoirs. This is the task:

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- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(f) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(g) If the smoke or fumes go away, then you corrected the fault.
1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

2) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.

3) Stop the left and right engines. This is the task:
- Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.

4) Stop the APU. This is the task:
- APU Usual Shutdown, AMM TASK 49-11-00-860-802.

D. Fault Isolation Procedure (Engine 2)

(1) Go to the FIM task for the observed fault: "Smoke or fumes in cabin: air supplied by engine 2" and do the steps in the task.

(a) Start the APU. This is the task:
- APU Starting and Operation, AMM TASK 49-11-00-860-801.

(b) Start the left and right engines. This is the task:
- Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
- Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

(d) Set these switches on the forward overhead panel to the specified positions:
1) ISOLATION VALVE switch to the OPEN position.

2) BLEED 1 and BLEED 2 switches to the ON position.

3) BLEED APU switch to the ON position.

(e) Pressurize the hydraulic reservoirs. This is the task:
- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(f) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(g) If the smoke or fumes go away, then you corrected the fault.
1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

2) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.

3) Stop the left and right engines. This is the task:
- Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.

4) Stop the APU. This is the task:
- APU Usual Shutdown, AMM TASK 49-11-00-860-802.

E. Fault Isolation Procedure (Oil Fumes)

(1) Go to the FIM task for these observed faults: "Smoke or fumes in cabin: air supplied by APU", "Smoke or fumes in cabin: air supplied by engine 1", "Smoke or fumes in cabin: air supplied by engine 2" and do the steps in the tasks.

(a) Start the APU. This is the task:
- APU Starting and Operation, AMM TASK 49-11-00-860-801.

(b) Start the left and right engines. This is the task:
- Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
- Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

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(d) Set these switches on the forward overhead panel to the specified positions:
1) ISOLATION VALVE switch to the OPEN position.
2) BLEED 1 and BLEED 2 switches to the ON position.
3) BLEED APU switch to the ON position.

(e) Pressurize the hydraulic reservoirs. This is the task:
- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(f) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(g) If the smoke or fumes go away, then you corrected the fault.
1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

2) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
3) Stop the left and right engines. This is the task:
- Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.

4) Stop the APU. This is the task:
- APU Usual Shutdown, AMM TASK 49-11-00-860-802.

F. Fault Isolation Procedure (Hydraulic Fluid)

(1) Do this check of the hydraulic reservoir pressurization module.

(a) Depressurize the hydraulic reservoirs. This is the task:
- Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802.

(b) Start the APU. This is the task:
- APU Starting and Operation, AMM TASK 49-11-00-860-801.

(c) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(d) Pressurize the hydraulic system reservoir with the pneumatic system pressurized. To pressurize, do the ?Reservoir Pressurization with the Pneumatic System Pressurized (Method 1) part of this task:
- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(e) Operate the APU and the cooling packs for 5 minutes.

(f) Remove conditioned air supplied by the left and right cooling packs. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(g) Set the BLEED APU switch to the OFF position.
1) Wait a minimum of 5 minutes to allow hydraulic fluid to leak past the check valve if there is a problem.

(h) Set the BLEED APU switch to the ON position.

(i) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(j) If the smoke or fumes increase or if they stay after five minutes, then do these steps:
1) Replace the hydraulic pressurization module. These are the tasks:
- Hydraulic Systems A and B Pressure Module Removal, AMM TASK 29-11-71-000-801,
- Hydraulic Systems A and B Pressure Module Installation, AMM TASK 29-11-71-400-801.

2) Start the APU. This is the task:
- APU Starting and Operation, AMM TASK 49-11-00-860-801.

3) Start the left and right engines. This is the task:
- Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.

4) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
- Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

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5) Set these switches on the forward overhead panel to the specified positions:
   a) ISOLATION VALVE switch to the OPEN position.
   b) BLEED 1 and BLEED 2 switches to the ON position.
   c) BLEED APU switch to the ON position.
6) Pressurize the hydraulic reservoirs. This is the task:
   • Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.
7) Supply conditioned air with the left and right cooling packs. This is the task:
   • Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
8) If the smoke or fumes go away, then you corrected the fault.
   a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
   b) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
   c) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
   d) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.

(k) If the smoke or fumes do not go away, then continue.

(2) Do this check of the air charging valve manifold assemblies.

(a) Depressurize the hydraulic reservoirs. This is the task:
   • Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802.

(b) Start the APU. This is the task:
   • APU Starting and Operation, AMM TASK 49-11-00-860-801.

(c) Supply conditioned air with the left and right cooling packs. This is the task:
   • Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(d) Pressurize the hydraulic system reservoir with an external air source. To pressurize, do the "Reservoir Pressurization with an External Air Source (Method 2)" part of this task:
   • Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(e) Operate the APU and the cooling packs for 5 minutes.

(f) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

(g) Set the BLEED APU switch to the OFF position.
   1) Wait a minimum of 5 minutes to allow hydraulic fluid to leak past the check valve if there is a problem.

(h) Set the BLEED APU switch to the ON position.

(i) Supply conditioned air with the left and right cooling packs. This is the task:
   • Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

(j) If the smoke or fumes increase or if they stay after five minutes, then do these steps:
   1) Replace the air charging valve manifold assembly. These are the tasks:
      • Air Charging Valve Manifold Assembly - Removal, AMM TASK 29-09-02-000-801,
      • Air Charging Valve Manifold Assembly - Installation, AMM TASK 29-09-02-400-801.
   2) Start the APU. This is the task:
      • APU Starting and Operation, AMM TASK 49-11-00-860-801.
   3) Start the left and right engines. This is the task:
      • Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.
   4) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
      • Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

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5) Set these switches on the forward overhead panel to the specified positions:
   a) ISOLATION VALVE switch to the OPEN position.
   b) BLEED 1 and BLEED 2 switches to the ON position.
   c) BLEED APU switch to the ON position.
6) Pressurize the hydraulic reservoirs. This is the task:
   • Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.
7) Supply conditioned air with the left and right cooling packs. This is the task:
   • Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
8) If the smoke or fumes go away, then you corrected the fault.
   a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
   b) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
   c) Stop the left and right engines. This is the task:
   • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
   d) Stop the APU. This is the task:
   • APU Usual Shutdown, AMM TASK 49-11-00-860-802.

G. Fault Isolation Procedure (Fuel Fumes)

(1) Go to the FIM task for these observed faults: "Smoke or fumes in cabin: air supplied by engine 1", "Smoke or fumes in cabin: air supplied by engine 2" and do the steps in the tasks.
   (a) Start the APU. This is the task:
   • APU Starting and Operation, AMM TASK 49-11-00-860-801.
   (b) Start the left and right engines. This is the task:
   • Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.
   (c) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
   • Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.
   (d) Set these switches on the forward overhead panel to the specified positions:
      1) ISOLATION VALVE switch to the OPEN position.
      2) BLEED 1 and BLEED 2 switches to the ON position.
      3) BLEED APU switch to the ON position.
   (e) Pressurize the hydraulic reservoirs. This is the task:
   • Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.
   (f) Supply conditioned air with the left and right cooling packs. This is the task:
   • Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
   (g) If the smoke or fumes go away, then you corrected the fault.
      1) Remove conditioned air supplied by the left and right cooling packs. This is the task:
      • Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.
      2) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
      3) Stop the left and right engines. This is the task:
      • Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.
      4) Stop the APU. This is the task:
      • APU Usual Shutdown, AMM TASK 49-11-00-860-802.

H. Fault Isolation Procedure (Electrical Faults)

(1) Look in OMF Existing Faults for maintenance messages related to a component that could be a source of the smoke or fumes.
   (a) If you find related maintenance messages, then do these steps:
      1) Do the fault isolation task for each related maintenance message.
      2) Start the APU. This is the task:

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- APU Starting and Operation, AMM TASK 49-11-00-860-801.

3) Start the left and right engines. This is the task:
   - Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.
4) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
   - Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.
5) Set these switches on the forward overhead panel to the specified positions:
   a) ISOLATION VALVE switch to the OPEN position.
   b) BLEED 1 and BLEED 2 switches to the ON position.
   c) BLEED APU switch to the ON position.
6) Pressurize the hydraulic reservoirs. This is the task:
   - Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.
7) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.
8) Supply conditioned air with the left and right cooling packs. This is the task:
   - Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.
9) If the smoke or fumes go away, then you corrected the fault.
   a) Remove conditioned air supplied by the left and right cooling packs. This is the task:
   - Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

   b) Stop the left and right engines. This is the task:
   - Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.

   c) Stop the APU. This is the task:
   - APU Usual Shutdown, AMM TASK 49-11-00-860-802.

(b) If you do not find any related maintenance message, then continue.

(2) Do this check of the cabin equipment.
   (a) Operate the cabin equipment, galley equipment, IFE equipment and light ballasts.
   (b) Operate one LRU at a time to isolate the source of smoke or fumes.
   (c) If you find smoke or fumes when the LRU operates, then do these steps:
      1) Clean or replace the LRU.
      2) Operate the LRU for a minimum of 3 minutes.
      3) If the smoke or fumes go away, then you corrected the fault.
   (d) If you do not find smoke or fumes when the LRU operates, then continue.

(3) Do this check of the recirculation system.

NOTE: Check the air recirculation fans and filters only if there is smoke or fumes in the passenger compartment.

   (a) Do this inspection of the air recirculation filters.
      1) Get access to the air recirculation filters. To get access, do this task:
         - Recirculation Air Filter - Removal, AMM TASK 21-25-01-000-801.
      2) If you find a problem with the filters, then do these steps:
         a) Install new air recirculation filters. This is the task:
         - Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.
      3) If you do not find a problem with the filters, then continue.
         a) Re-install the air recirculation filters. This is the task:
         - Recirculation Air Filter - Installation, AMM TASK 21-25-01-400-801.
   (b) Do this check of the air recirculation fans and the related wiring. This is the task:
      - Recirculation System - Operational Test, AMM TASK 21-25-00-710-801.
      1) If the smoke or fumes increase or if they remain after five minutes, then do these steps:
         a) Replace the applicable air recirculation fan. These are the tasks:
         - Recirculation Fan Removal, AMM TASK 21-25-02-000-801,

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- Recirculation Fan Installation, AMM TASK 21-25-02-400-801.

b) Repair the applicable wiring.

c) Start the APU. This is the task:
- APU Starting and Operation, AMM TASK 49-11-00-860-801.

d) Start the left and right engines. This is the task:
- Start the Engine (Selection), AMM TASK G71-00-00-910-802-G00.

e) Make sure that you obey all safety precautions when you are near an engine that is in operation. The safety precautions are in this task:
- Engine Ground Safety Precautions, AMM TASK G71-00-00-800-801-G00.

f) Set these switches on the forward overhead panel to the specified positions:
<1> ISOLATION VALVE switch to the OPEN position.
<2> BLEED 1 and BLEED 2 switches to the ON position.
<3> BLEED APU switch to the ON position.

g) Pressurize the hydraulic reservoirs. This is the task:
- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

h) Set the L RECIRC FAN and R RECIRC FAN switches to the AUTO position.

i) Supply conditioned air with the left and right cooling packs. This is the task:
- Supply Conditioned Air with a Cooling Pack, AMM TASK 21-00-00-800-803.

j) If the smoke or fumes go away, then you corrected the fault.
<1> Remove conditioned air supplied by the left and right cooling packs. This is the task:
- Remove Conditioned Air Supplied by a Cooling Pack, AMM TASK 21-00-00-800-804.

<2> Stop the left and right engines. This is the task:
- Stop the Engine (Usual Engine Stop), AMM TASK G71-00-00-910-806-G00.

<3> Stop the APU. This is the task:
- APU Usual Shutdown, AMM TASK 49-11-00-860-802.

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